02_read_committed_vs_rcsi

Propietario	© Carlos Roberto
Etiquetas	

2.1 - DEFAULT BEHAVIOR (Read Committed, no RCSI)

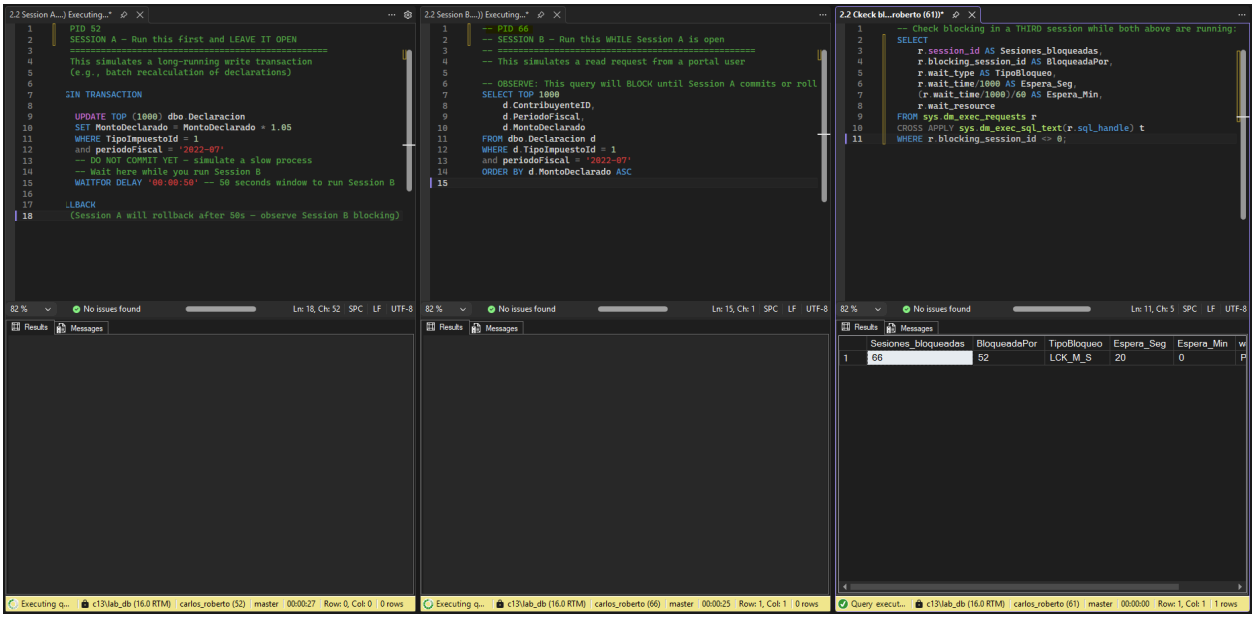
```
USE STN_Lab;
GO

-- ============================================================
==
-- PART 1 — DEFAULT BEHAVIOR (Read Committed, no RCSI)
-- ============================================================
==

-- Verify RCSI is OFF before starting
SELECT
    name,
    CASE WHEN is_read_committed_snapshot_on = 0 THEN 'OFF' EL
SE 'ON' END AS RCSI_Enabled
FROM sys.databases
WHERE name = 'STN_Lab';
-- Expected: RCSI_Enabled = OFF
GO
```

2.2 Simulation long-running write transaction

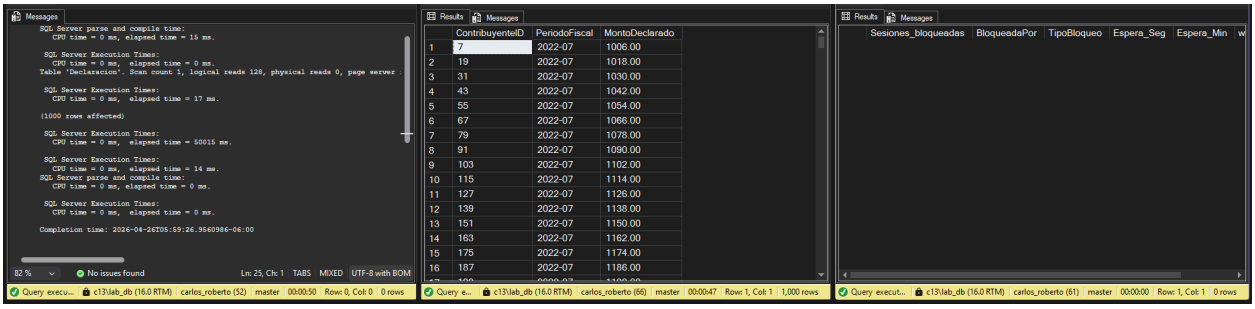
```
-- PID 52
-- SESSION A — Run this first and LEAVE IT OPEN
-- ==================================================
-- This simulates a long-running write transaction
```

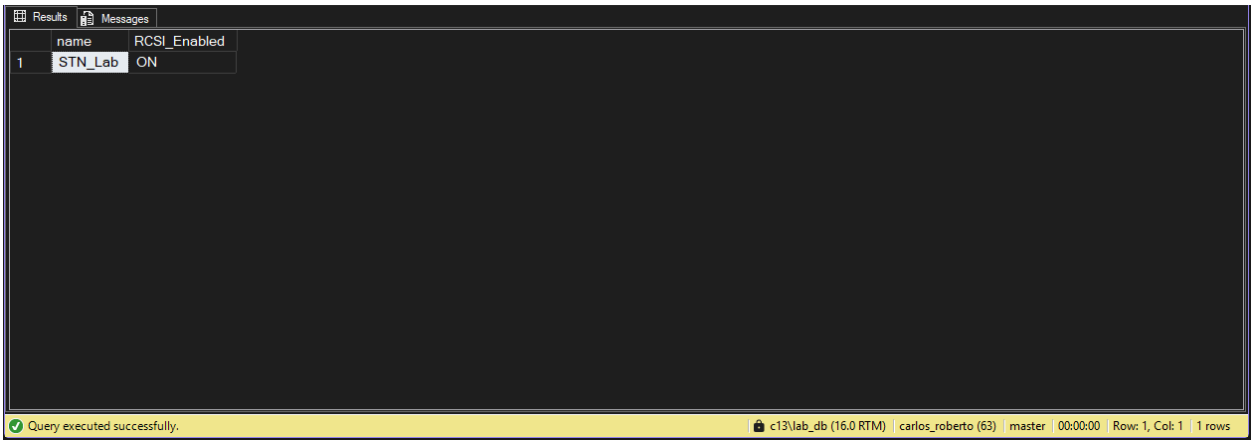
```sql
-- (e.g., batch recalculation of declarations)

BEGIN TRANSACTION

    UPDATE TOP (1000) dbo.Declaracion
    SET MontoDeclarado = MontoDeclarado * 1.05
    WHERE TipoImpuestoId = 1
    and periodoFiscal = '2022-07'
    -- DO NOT COMMIT YET — simulate a slow process
    -- Wait here while you run Session B
    WAITFOR DELAY '00:00:50' -- 50 seconds window to run Sess
ion B

ROLLBACK
-- (Session A will rollback after 50s — observe Session B blo
cking)

-- PID 66
-- ====================================================
-- SESSION B — Run this WHILE Session A is open
-- ====================================================
-- This simulates a read request from a portal user

-- OBSERVE: This query will BLOCK until Session A commits or
rolls back
SELECT TOP 1000
    d.ContribuyenteID,
    d.PeriodoFiscal,
    d.MontoDeclarado
FROM dbo.Declaracion d
WHERE d.TipoImpuestoId = 1
and periodoFiscal = '2022-07'
ORDER BY d.MontoDeclarado ASC


-- Check blocking in a THIRD session while both above are run
```

```
ning:
SELECT
    r.session_id AS Sesiones_bloqueadas,
    r.blocking_session_id AS BloqueadaPor,
    r.wait_type AS TipoBloqueo,
    r.wait_time/1000 AS Espera_Seg,
    (r.wait_time/1000)/60 AS Espera_Min,
    r.wait_resource
FROM sys.dm_exec_requests r
CROSS APPLY sys.dm_exec_sql_text(r.sql_handle) t
WHERE r.blocking_session_id <> 0;
```



Session 52 is running its update, while session 66 only wants to read data; there are locks for session 66. Wait for 50 seconds for the results.



2.3 ENABLE RCSI (eliminates blocking for readers)

```sql
ALTER DATABASE STN_Lab SET READ_COMMITTED_SNAPSHOT ON WITH RO
LLBACK IMMEDIATE;
GO

-- Verify RCSI is now ON
SELECT
    name,
    CASE WHEN is_read_committed_snapshot_on = 0 THEN 'OFF' EL
SE 'ON' END AS RCSI_Enabled
FROM sys.databases
WHERE name = 'STN_Lab';
-- Expected: RCSI_Enabled = ON
GO
```